Section 2.3: Lines and Planes (Reference: Section 1.3)

1) Equation of a line l in $\mathbb{R}^2$:

a) General form:

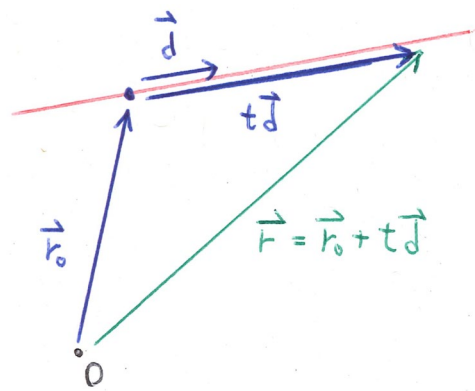
(19a)
$$ax + by = c$$

Note: With a, b, and c sharing no common factor, general form of a line in $\mathbb{R}^2$ <u>is unique</u> (up to a sign change).

b) Vector form:

(20a)
$$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{d}} \quad (-\infty < t < \infty)$$

where $\vec{\mathbf{r}}_0 = [x_0 \quad y_0]$ is the standard position vector for any specific point on the line l and $\vec{\mathbf{d}} = [d_1 \quad d_2]$ is <u>a direction vector</u> for l. As t increases, we travel on the line in the direction of $\vec{\mathbf{d}}$.



Note: Since any nonzero scalar multiple of $\vec{\mathbf{d}}$ is still a direction vector for the line and there are many points on the line that we can use to determine $\vec{\mathbf{r}}_0$, vector form of a line in $\mathbb{R}^2$ <u>is not unique</u>.

c) The equations corresponding to the components of the vector form of the equation are called <u>parametric equations</u> of l:

(21a)
$$x = x_0 + d_1 t, \quad y = y_0 + d_2 t$$

Note: Since vector form of a line l is not unique, parametric equations of l in $\mathbb{R}^2$are also <u>not unique</u>.

Example 2.3.1. Find parametric equations of the line passing through the points $P(2, -1)$ and $Q(10, 3)$.

Solution: Using the vector form with $\vec{\mathbf{r}}_0 = [2 \quad -1]$ and direction vector $\vec{\mathbf{d}} = \overrightarrow{PQ} = [10 - 2 \quad 3 - 1 = [8 \quad 4]$, we have

$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{d}} \Longrightarrow [x \quad y] = [2 \quad -1] + t[8 \quad 4]$

$\Longrightarrow [x \quad y] = [2 + 8t \quad -1 + 4t] \Longrightarrow \boxed{\begin{array}{l} \boldsymbol{x = 2 + 8t} \\ \boldsymbol{y = -1 + 4t} \end{array}}$

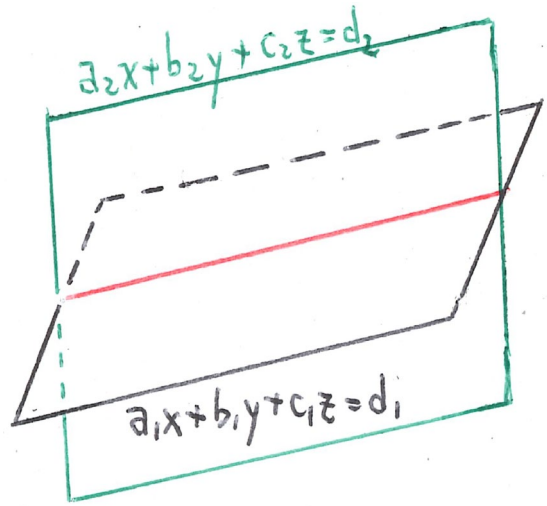
Note: There are many correct answers (as we can use any point on the line for $\vec{\mathbf{r}}_0$ and any non-zero vector parallel to the line for $\vec{\mathbf{d}}$).

2) Equation of a line l in $\mathbb{R}^3$:

a) General form:

(19b) $$a_1x + b_1y + c_1z = d_1 \quad \text{and} \quad a_2x + b_2y + c_2z = d_2$$

We can interpret this general form as follows: a line in $\mathbb{R}^3$ is the intersection of two planes. Since any line in $\mathbb{R}^3$ is the intersection of many pairs of planes, general form of a line in $\mathbb{R}^3$ <u>is not unique</u>.



b) For the line l in 3-space through the point $P_0(x_0, y_0, z_0)$ and parallel to the nonzero vector $\vec{\mathbf{d}} = [d_1 \quad d_2 \quad d_3]$, let $\vec{\mathbf{r}}_0 = [x_0 \quad y_0 \quad z_0]$ be the vector from the origin to P_0. For any point $P(x, y, z)$ on l, there is a scalar t $(-\infty < t < \infty)$ such that the vector from the origin to the point P, $\vec{\mathbf{r}} = [x \quad y \quad z]$, satisfies the vector equation

(20b) $$\vec{\mathbf{r}} - \vec{\mathbf{r}}_0 = t\vec{\mathbf{d}} \quad \Rightarrow \quad \vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{d}}$$

This is called the <u>vector form</u> of the equation of the line; $\vec{\mathbf{d}}$ is <u>a direction vector</u> for l As t increases, we travel on the line in the direction of $\vec{\mathbf{d}}$.

Note: Just like vector form of a line in $\mathbb{R}^2$, since any nonzero scalar multiple of $\vec{\mathbf{d}}$ is still a direction vector for the line and there are many points on the line that we can use to determine $\vec{\mathbf{r}}_0$, vector form of a line in $\mathbb{R}^3$ <u>is not unique</u>.

c) The equations corresponding to the components of the vector form of the equation are called <u>parametric equations</u> of l:

(21b) $$x = x_0 + d_1t, \quad y = y_0 + d_2t, \quad z = z_0 + d_3t$$

Note: Since vector form of a line l is not unique, parametric equations of l in $\mathbb{R}^3$ are also <u>not unique</u>.

Example 2.3.2. Find parametric equations of the line passing through the points $P(-9,3,4)$ and $Q(-6,-15,4)$.

Solution: Using the vector form with $\vec{\mathbf{r}}_0 = [-9 \quad 3 \quad 4]$ and direction vector $\vec{\mathbf{d}} = \overrightarrow{PQ} = [(-6)-(-9) \quad (-15)-3 \quad 4-4] = [3 \quad -18 \quad 0]$, we have $\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{d}} \Longrightarrow [x \quad y \quad z] = [-9 \quad 3 \quad 4] + t[3 \quad -18 \quad 0]$

$$\Longrightarrow [x \quad y \quad z] = [-9+3t \quad 3-18t \quad 4] \Longrightarrow \boxed{\begin{aligned} x &= -9+3t \\ y &= 3-18t \\ z &= 4 \end{aligned}}$$

Note: There are many correct answers (as we can use any point on the line for $\vec{\mathbf{r}}_0$ and any non-zero vector parallel to the line for $\vec{\mathbf{d}}$).

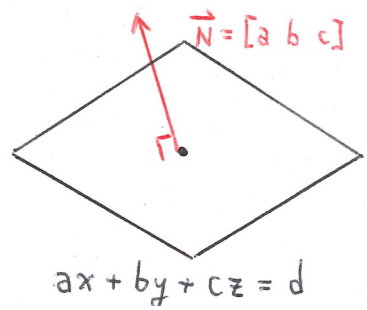
3) Equation of a plane in $\mathbb{R}^3$:

a) The general form:

(19c) $$ax + by + cz = d$$

where $\vec{\mathbf{N}} = [\mathrm{a} \quad \mathrm{b} \quad \mathrm{c}]$ is a vector normal to the plane

Note: With a, b, c and d sharing no common factor, general form of a plane in R^3 <u>is unique</u> (up to a sign change).

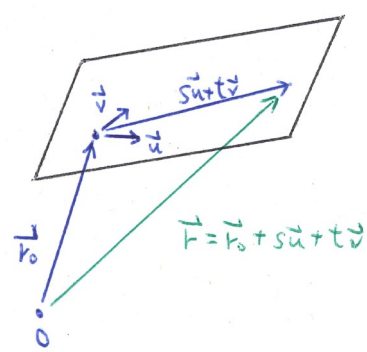


b) Vector form:

(20c) $$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + s\vec{\mathbf{u}} + t\vec{\mathbf{v}}$$

where $\vec{\mathbf{r}}_0 = [x_0 \quad y_0 \quad z_0]$ is a point on the plane and $\vec{\mathbf{u}} = [u_1 \quad u_2 \quad u_3]$ and $\vec{\mathbf{v}} = [v_1 \quad v_2 \quad v_3]$ are any two <u>direction vectors</u> for the plane (i.e. $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are non-zero vectors parallel to P but not parallel to each other)

Note: Since there are many choices for the direction vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ as well as the point to determine $\vec{\mathbf{r}}_0$, vector form of a plane in $\mathbb{R}^3$ <u>is not unique</u>.



c) The equations corresponding to the components of the vector form of the equation are called parametric equations of P.

(21c) $$x = x_0 + u_1 s + v_1 t, \quad y = y_0 + u_2 s + v_2 t, \quad z = z_0 + u_3 s + v_3 t$$

Note: Since vector form of a plane is not unique, parametric equations of a plane P in $\mathbb{R}^3$ are also <u>not unique</u>.

Example 2.3.3. Find parametric equations of the plane passing through the points $P(5,0,0)$, $Q(6,1,0)$ and $R(3,0,1)$.

Solution: Using the vector form with $\vec{\mathbf{r}}_0 = [5 \quad 0 \quad 0]$ and direction vectors $\vec{\mathbf{u}} = \overrightarrow{PQ} = [6-5 \quad 1-0 \quad 0-0] = [1 \quad 1 \quad 0]$ and $\vec{\mathbf{v}} = \overrightarrow{PR} = [3-5 \quad 0-0 \quad 1-0] = [-2 \quad 0 \quad 1]$ we have

$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + s\vec{\mathbf{u}} + t\vec{\mathbf{v}} \Longrightarrow [x \quad y \quad z] = [5 \quad 0 \quad 0] + s[1 \quad 1 \quad 0] + t[-2 \quad 0 \quad 1]$

$\Longrightarrow [x \quad y \quad z] = [5+s-2t \quad s \quad t] \Longrightarrow \boxed{\begin{aligned} x &= 5+s-2t \\ y &= s \\ z &= t \end{aligned}}$

Note: There are many correct answers (as we can use any point on the plane for $\vec{\mathbf{r}}_0$ and any two non-zero non-parallel vectors parallel to the plane for $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$).

Practice Exercises

1. (Assignment 2) (a) Find parametric equations for the line passing through the points $A(2,4,-1)$ and $B(5,0,7)$.
   [Note: There are many correct answers, such as $x = 2+3t, y = 4-4t, z = -1+8t$]
   (b) At which point does the line intersect the xy-plane?
   [Answer: $\left(\frac{19}{8}, \frac{7}{2}, 0\right)$]

2. Find the distance between the two lines.
   (a) $\begin{Bmatrix} x = 4-s, & y = 3+s, & z = 4s-1 \\ x = t-6, & y = 2t+13, & z = 2t+12 \end{Bmatrix}$
   (b) $\begin{Bmatrix} x = 4s+2, & y = 4-s, & z = s-3 \\ x = 2-12t, & y = 3t, & z = 2-3t \end{Bmatrix}$
   [Answer: (a) (Non-Parallel Lines) 9 (b) (Parallel Lines) $\frac{1}{2}\sqrt{146}(\approx 6.0415)$]

3. (Assignment 2) Find parametric equations of the plane passing through the points $P(5,4,7), P, Q(0,-8,2)$, and $R(1,2,-8)$.
   [Note: There are many correct answers, such as
   $x = 5-5s-4t, y = 4-12s-2t, z = 7-5s-15t$]

4. Find an equation in general form for the plane passing through the points $A(0,1,0)$, $B(3,2,0)$ and $C(0,3,-3)$.
   [Answer: $-x+3y+2z = 3$]

5. Find the distance between the point $P = (2,-3,7)$ and the plane $4x-7y+11z = 12$.
   [Answer: $\frac{47}{93}\sqrt{186}$]